

3. Project Design

Project Name

ComicCraft - AI Comic Story Creator using Gemini Models

1. Introduction

The Project Design describes the structure, components, workflow, and interface of the ComicCraft application. The system is designed to generate creative comic stories using Google's Gemini AI models.

2. System Architecture

The ComicCraft system consists of the following main components:

- User Interface – Allows users to enter story ideas and character details.
- Flask Backend – Handles user requests and application logic.
- Gemini AI – Generates story content, characters, scenes, and dialogues.
- Comic Preview – Displays the generated comic story to the user.

System Flow

User | v Web Interface | v Flask Backend | v Gemini AI Model | v Generated Story | v Comic Preview

3. Module Design

3.1 User Input Module

This module collects the user's:

- Story idea
- Character details
- Story preferences
- Other required inputs

3.2 AI Story Generation Module

This module sends the user's input to the Gemini AI model and receives the generated story content.

3.3 Character and Scene Module

This module organizes the generated content into:

- Characters
- Scenes
- Dialogues
- Story sequence

3.4 Comic Preview Module

This module displays the generated comic content in a user-friendly format.

3.5 Backend Module

The Flask backend manages:

- User requests
 - API communication
 - Data processing
 - Response handling
 - Page navigation
4. User Interface Design

The application can contain the following pages:

1. Home Page
2. Story Creation Page
3. Character Input Page
4. Comic Generation Page
5. Comic Preview Page

The interface is designed to be simple and easy for beginners to use.

5. Data Flow

User Input | v Input Validation | v Flask Backend | v Gemini API | v AI Generated Content | v
Process Story | v Display Comic Preview

6. Technology Design

Component| Technology Frontend| HTML, CSS, JavaScript Backend| Python Flask AI|
Google Gemini Development Tool| Visual Studio Code Version Control| GitHub

7. Folder Structure

```
ComicCraft AI/ | ├── app.py ├── routes.py ├── requirements.txt | ├── templates/ |  
| ├── index.html | └── comic_preview.html | ├── static/ | └── style.css | ├──  
1_Brainstorming_and_Ideation/ | └── README.md | ├── 2_Requirement_Analysis/ | └──  
README.md | └── 3_Project_Design/ └── README.md
```

8. Design Goals

- Simple user interface
- Easy story creation
- AI-powered content generation
- Fast and organized workflow
- Beginner-friendly experience
- Clear comic preview

9. Expected Design Outcome

The project design provides a clear structure for developing ComicCraft. It connects the user interface, Flask backend, Gemini AI, and comic preview into a single workflow.

10. Conclusion

The Project Design serves as a blueprint for implementing ComicCraft. It defines the system architecture, modules, data flow, interface structure, and technologies required to build the application.